

Stack Operations Using Python – Menu Driven Program

The following **menu-driven Python program** implements all basic stack operations using an array (list):

- **1. PUSH** – Insert an element
- **2. POP** – Delete the top element
- **3. PEEK** – Display the top element
- **4. DISPLAY** – Display all stack elements
- **5. isEmpty** – Check whether stack is empty
- **6. isFull** – Check whether stack is full
- **7. EXIT** – Exit the program

Menu Driven Program for Stack Operations

MAX = 5

stack = []

top = -1 // Here, top is a **global variable**.

PUSH Operation

def push():

 global top

 // Without global top, Python treats top inside the function as a **local variable**, causing an error

 //when we try to use its previous value.

 if top == MAX - 1:

 print("Stack Overflow! Stack is Full.")

 else:

 item = int(input("Enter element to push: "))

 top = top + 1

 stack.append(item)

 print(item, "inserted into the stack.")

POP Operation

```
def pop():  
    global top  
    if top == -1:  
        print("Stack Underflow! Stack is Empty.")  
    else:  
        item = stack.pop()  
        top = top - 1  
        print(item, "deleted from the stack.")
```

PEEK Operation

```
def peek():  
    if top == -1:  
        print("Stack is Empty.")  
    else:  
        print("Top element is:", stack[top])
```

DISPLAY Operation

```
def display():  
    if top == -1:  
        print("Stack is Empty.")  
    else:  
        print("Stack elements are:")  
  
        for i in range(top, -1, -1):  
            print(stack[i])
```

isEmpty Operation

def is_empty():

if top == -1:

print("Stack is Empty.")

else:

print("Stack is Not Empty.")

isFull Operation

def is_full():

if top == MAX - 1:

print("Stack is Full.")

else:

print("Stack is Not Full.")

print("Available space:", MAX - 1 - top)

Main Menu

while True:

print("\n===== STACK MENU =====")

print("1. PUSH")

print("2. POP")

print("3. PEEK")

print("4. DISPLAY")

print("5. isEmpty")

print("6. isFull")

print("7. EXIT")

print("=====")

choice = int(input("Enter your choice: "))

```
if choice == 1:
```

```
    push()
```

```
elif choice == 2:
```

```
    pop()
```

```
elif choice == 3:
```

```
    peek()
```

```
elif choice == 4:
```

```
    display()
```

```
elif choice == 5:
```

```
    is_empty()
```

```
elif choice == 6:
```

```
    is_full()
```

```
elif choice == 7:
```

```
    print("Program terminated.")
```

```
    break
```

```
else:
```

```
    print("Invalid choice! Please try again.")
```